Sagar Kriplani

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PROFESSIONAL PROFILE

Results-driven Mechanical Design & Development Engineer (M.Sc. Scientific Instrumentation, B.E. Mechanical Engineering) and AI enthusiast with over 3 years of hands-on industry experience across custom machinery (Sondermaschinenbau), semiconductor equipment, and thermomechanical FEA simulation. Pursuing continuous professional development in CAD and FEM AI automation workflows and open-source tooling on GitHub (FreeCAD, Python). Proven concept-to-release track record in 3D CAD modeling (Autodesk Inventor 2024, PTC Creo, SolidWorks, Siemens NX), DIN EN ISO 1101 GD&T drawing creation, and DFM/DFA design yielding up to 70% cost reduction and 60% weight optimization. Engineered high-vacuum and thermal modules per SEMI S2/S8 standards at Brooks Automation; led design release for 8+ custom platforms at NeSt GmbH. Interactive engineering portfolio available at sagar-kriplani.github.io. German B2, English C1. Unrestricted EU Blue Card, immediately available.

WORK EXPERIENCE

09.2025 - 09.2026

Mechanical Design & Development Engineer

NeSt GmbH Sondermaschinenbau & Anlagenbau
Herbrechtingen, Germany

- Managed full concept-to-release design cycles for over 8 custom machine platforms, generating approx. 200 production drawings per month compliant with DIN/ISO standards.
- Engineered complex machine frames, sheet metal enclosures with Trumpf unfolding, and welded assemblies applying strict DFM/DFA principles.
- Cut component manufacturing costs by up to 70% and reduced structural weight by 60% through standardisation, material substitution, and FMEA-guided revisions.
- Maintained revision-controlled CAD assemblies and Bills of Materials (BOM) in PDM and SAGE ERP; coordinated technical interfaces across R&D, assembly, and suppliers.
- Performed coupled structural and thermal FEA simulations in ANSYS Workbench alongside DIN EN ISO 1101 GD&T tolerance stack calculations to validate mechanical integrity.

01.2023 - 06.2024

Mechanical Design & Development Engineer (Semiconductor & Thermal Systems)

Brooks Automation (Deutschland) GmbH
Steißlingen, Germany

- Redesigned the Process Air Unit (PAU) for EUV Pod Cleaners from concept through cleanroom prototype validation; implemented stainless steel 304/316Ti piping per ASME B31.3.
- Cut electrical heating power from 48 kW to 18 kW (~50% design energy savings), reduced pressure loss by 90% (500 to 50 mbar), and lowered physical footprint by 34%.
- Engineered high-vacuum chamber modules and mechatronic assemblies for FOUP/EUV cleaning systems per SEMI S2/S8; maintained CAD models, BOMs, and ECNs in Agile PLM.
- Evaluation of IR LEDs in the wet cleaning chamber: Conceptual design, optothermal evaluation, and experimental testing of high-power IR LED arrays for targeted drying and polymer heating under cleanroom class 1 µm conditions.
- Re-engineered IR quartz heater mounting brackets with 80% reduced thermal shadowing and 3x weight reduction; validated via ANSYS FEA simulation and FLIR thermography with ±2 °C temperature uniformity (Master's thesis, Grade 1.3).

08.2020 - 08.2021

Design & Development Engineer

Ravi Brass Industries
Jamnagar, India

- Engineered precision electrical connectors and micro-components from brass and copper alloys using PTC Creo per DIN/IEC standards.
- Designed 4 forming and injection molding tools for high-volume series production exceeding 100,000 components/month; cut scrap by 20% via tolerance optimization.

PROJECTS

10.2026 - Present

CAD Automation & Simulation Interoperability Hub (FreeCAD)

Open-Source Project on GitHub

- Developed parametric 3D modeling scripts in Python for FreeCAD as well as automated CAD conversion and export pipelines (STEP/STL).

08.2022

Automated Handling & Manipulation Tool (Robotic Arm)

Ernst-Abbe-Hochschule Jena | Grade 1.3

- Engineered a high-precision robotic arm mechanism in Autodesk Inventor for transferring parts between conveyor belts with ±50 µm positioning accuracy.

03.2022 - 08.2022

Engineering Machine Learning & Simulation Hub

Ernst-Abbe-Hochschule Jena | Subject Project, Grade 2.0

- Developed automated data analysis pipelines in Python (NumPy, SciPy, Scikit-Learn) for physical measurement series and FTIR spectroscopy data.

05.2020 - 07.2020

Gearbox & Transmission Design for Agricultural Machinery

Smit Engineering | Agricultural Implements

- Engineered 3D CAD models of gearboxes, differentials, shafts, and bearing arrangements for rotavators; detailed housings and drawings per DIN/ISO.

EDUCATION

10.2021 - 03.2025

Master of Science (M.Sc.) in Scientific Instrumentation

Ernst-Abbe-Hochschule Jena, Germany

Final Grade: 1.8 (ECTS A).

Master's Thesis: Comparative study of radiant heaters and LEDs for polymer heating in FOUP cleaner vacuum chambers (Brooks Automation). Thermal system design, coupled ANSYS FEA, high-vacuum test validation; ±2 °C uniformity.

08.2017 - 08.2020

Bachelor of Engineering (B.E.) in Mechanical Engineering

Gujarat Technological University (ADIT), India

Final Grade: 1.9 (CGPA 8.14/10.00).

Bachelor's Thesis: Numerical and mathematical modeling of frost growth on a flat plate. MATLAB, CFD simulation of heat and mass transfer for cryogenic and HVAC systems.

08.2014 - 06.2017

Diploma in Mechanical Engineering

Gujarat Technological University (GP Jamnagar), India

Final Grade: 1.2 (CGPA 9.50/10.00). Focus on technical mechanics, thermodynamics, 3D CAD modeling, and manufacturing technologies.

TECHNICAL SKILLS & COMPETENCIES

3D CAD & Mechanical Design	Autodesk Inventor 2024 (Expert) · PTC Creo Parametric · SolidWorks · Siemens NX · AutoCAD; Production Drawings · DIN EN ISO 1101 (GD&T) · Tolerance Stacks
Manufacturing & DFM/DFA	Custom Machinery (Sondermaschinenbau) · Precision Engineering · Sheet Metal Design (Trumpf Unfolding) · Welded Assemblies · DFM/DFA · FMEA · Standardization
Simulation & Calculation	ANSYS Workbench (Coupled Structural & Thermal FEA · Modal Analysis) · CFD Fundamentals · MATLAB/Simulink · Heat & Mass Transfer · ASME B31.3
Semiconductor & Cleanroom	Cleanroom Class 1 µm · High-Vacuum & Thermal Subsystems · SEMI S2/S8 Standards · Optothermal Assemblies · FLIR Thermography · Wafer Handling
PLM, IT & Languages	Agile PLM · Autodesk Vault/PDM · SAGE ERP · BOM & ECN Management · Python (Fundamentals) · MS Office 365; German B2 · English C1; EU Blue Card

PROFESSIONAL DEVELOPMENT / CERTIFICATIONS

CAD & FEM AI Automation Workflows and M365 Copilot	Continuous Professional Development	10/2026 - Present
Semiconductor Innovation	Tech Diplomacy Academy	2024
ANSYS Basic to Professional Training (FEA Structural & Thermal)	ELEATION	2021
Siemens NX: Basic to NC Programming	Siemens Centre of Excellence	2020

Heidenheim an der Brenz, 05.11.2026

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